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## **FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)**

### **TD N°\_5 – RÉSEAUX**

**COURS: RÉSEAUX**

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**NOM:** PÉTIOTE

**NIVEAU:** L3

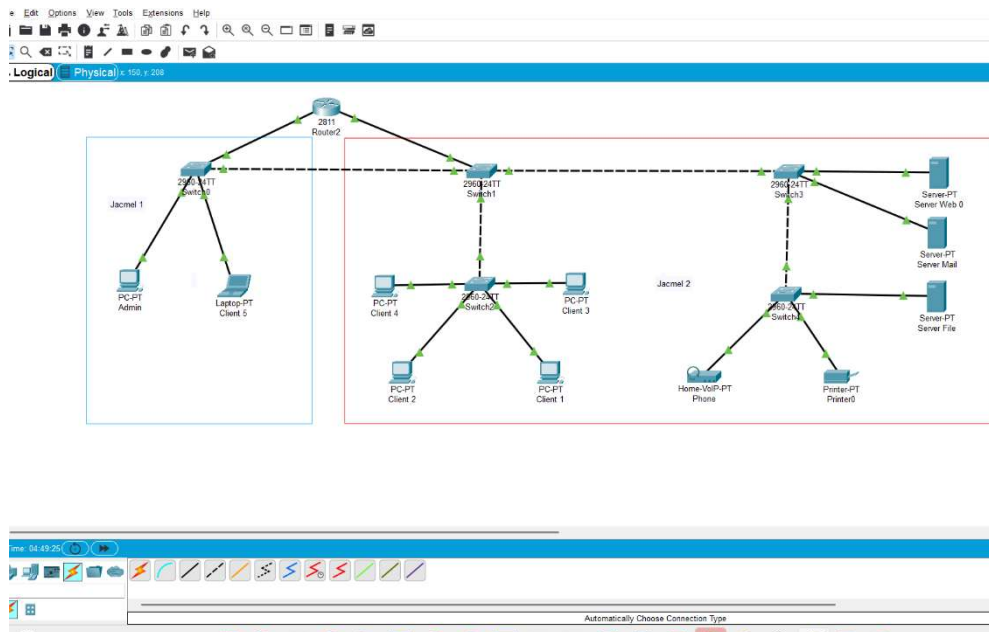
**DATE:** 27/11/2025

## L'objectif de ce TD est de:

1. Configurer un serveur DNS centralisé dans un réseau.
2. Associer plusieurs noms de domaine à différents serveurs (Web, Mail, FTP).
3. Intégrer le DNS avec d'autres services (Web Server).
4. Tester la résolution de noms depuis des PC clients et depuis le routeur.
5. Configurer un serveur Web (HTTP et HTTPS) dans Cisco Packet Tracer.
6. Comprendre la différence entre HTTP et HTTPS.
7. Tester l'accès aux pages web depuis des PC clients.

1. Reproduisez cette topologie en configurant les services DNS, HTTP et HTTPS, puis en affichant une page web.

a- Reproduction de la topologie.



a-1) Configuration du router, des switches et du server DNS.

```

J1>enable
J1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
J1(config)#ip DNS 192.168.2.10
      ^
% Invalid input detected at '^' marker.

J1(config)#ip name-server 192.168.2.10
J1(config)#end
J1#
%SYS-5-CONFIG_I: Configured from console by console
|

```

Ctrl+F6 to exit CLI focus

```

S3W1>enable
S3W1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S3W1(config)#interface Vlan 1
S3W1(config-if)#ip address 192.168.1.1 255.255.255.0
S3W1(config-if)#no shutdown
S3W1(config-if)#ip default-gateway 192.168.1.4
S3W1(config)#end
S3W1#
%SYS-5-CONFIG_I: Configured from console by console

```

trl+F6 to exit CLI focus

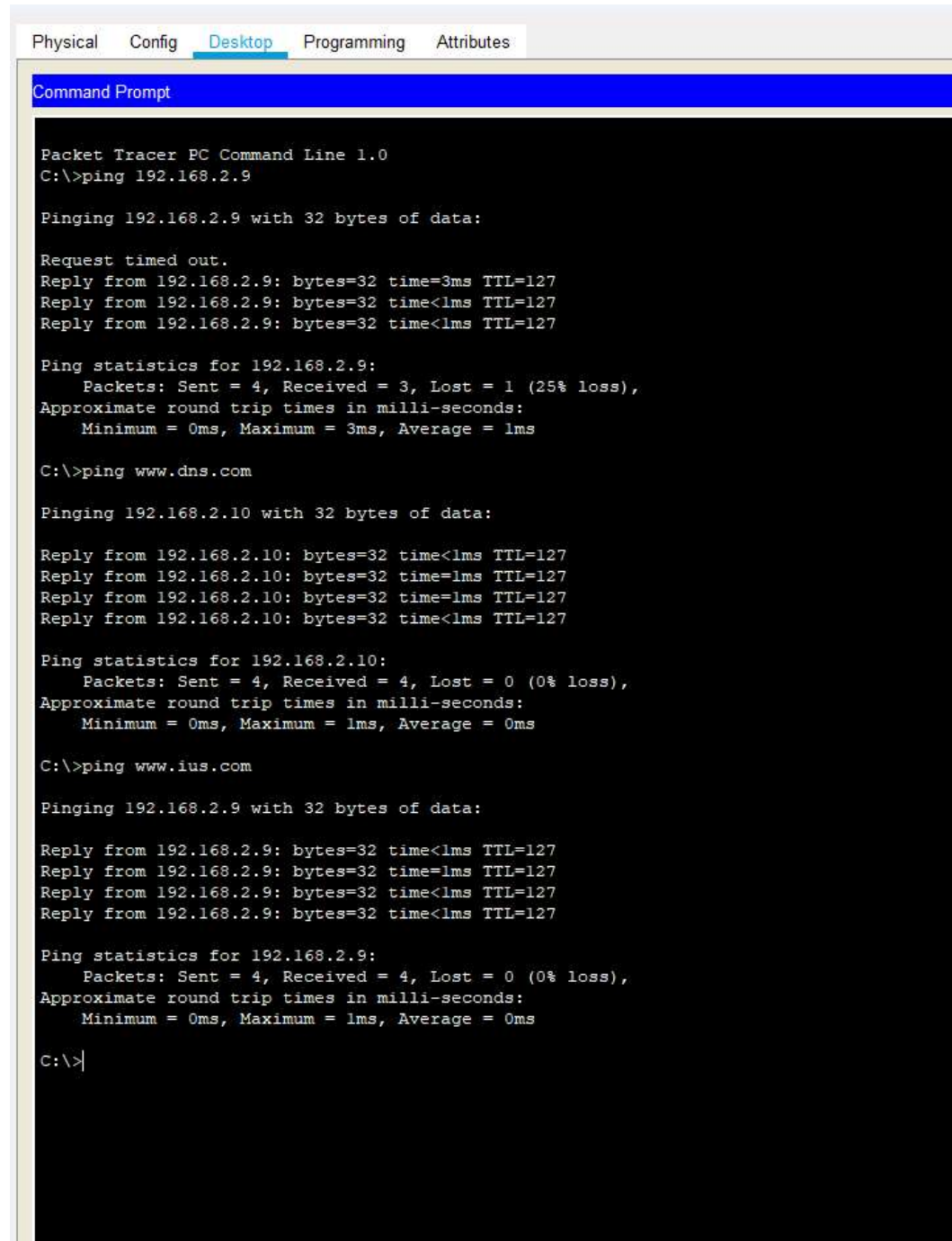
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Top

| Physical Config <u>Services</u> Desktop Programming Attributes                                                                                |                                                                           |             |               |              |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-------------|---------------|--------------|--------|
| <b>SERVICES</b><br>HTTP<br>DHCP<br>DHCPv6<br>TFTP<br><b>DNS</b><br>SYSLOG<br>AAA<br>NTP<br>EMAIL<br>FTP<br>IoT<br>VM Management<br>Radius EAP | DNS                                                                       |             |               |              |        |
|                                                                                                                                               | DNS Service <input checked="" type="radio"/> On <input type="radio"/> Off |             |               |              |        |
|                                                                                                                                               | Resource Records                                                          |             |               |              |        |
|                                                                                                                                               | Name                                                                      |             | Type A Record |              |        |
|                                                                                                                                               | Address                                                                   |             |               |              |        |
|                                                                                                                                               | Add                                                                       |             | Save          |              | Remove |
|                                                                                                                                               | No.                                                                       | Name        | Type          | Detail       |        |
|                                                                                                                                               | 0                                                                         | admin       | A Record      | 192.168.1.2  |        |
|                                                                                                                                               | 1                                                                         | client 1    | A Record      | 192.168.2.5  |        |
|                                                                                                                                               | 2                                                                         | client 2    | A Record      | 192.168.2.4  |        |
|                                                                                                                                               | 3                                                                         | client 3    | A Record      | 192.168.2.6  |        |
|                                                                                                                                               | 4                                                                         | client 4    | A Record      | 192.168.2.3  |        |
|                                                                                                                                               | 5                                                                         | client 5    | A Record      | 192.168.1.3  |        |
|                                                                                                                                               | 6                                                                         | server file | A Record      | 192.168.2.11 |        |
|                                                                                                                                               | 7                                                                         | www.dns.com | A Record      | 192.168.2.10 |        |
|                                                                                                                                               | 8                                                                         | www.ius.com | A Record      | 192.168.2.9  |        |

-Ping pour tester la connectivité entre les hôtes avec des noms de domaine.



The screenshot shows the Packet Tracer PC Command Line 1.0 interface. The 'Desktop' tab is selected. The command prompt displays the results of three ping tests. The first test is to 192.168.2.9, which shows a 25% loss. The second test is to www.dns.com (192.168.2.10), which shows 0% loss. The third test is to www.ius.com (192.168.2.9), which also shows 0% loss.

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.9

Pinging 192.168.2.9 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.9: bytes=32 time=3ms TTL=127
Reply from 192.168.2.9: bytes=32 time<1ms TTL=127
Reply from 192.168.2.9: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.9:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>ping www.dns.com

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time=1ms TTL=127
Reply from 192.168.2.10: bytes=32 time=1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping www.ius.com

Pinging 192.168.2.9 with 32 bytes of data:

Reply from 192.168.2.9: bytes=32 time<1ms TTL=127
Reply from 192.168.2.9: bytes=32 time=1ms TTL=127
Reply from 192.168.2.9: bytes=32 time<1ms TTL=127
Reply from 192.168.2.9: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

## Exemple d'un de nos PC

|                    |               |
|--------------------|---------------|
| IP Address         | 192.168.1.2   |
| Subnet Mask        | 255.255.255.0 |
| Default Gateway    | 192.168.1.4   |
| DNS Server         | 192.168.2.10  |
| IPv6 Configuration |               |

## b-Configuration HTTP et HTTPS

Physical Config **Services** Desktop Programming Attributes

**SERVICES**

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

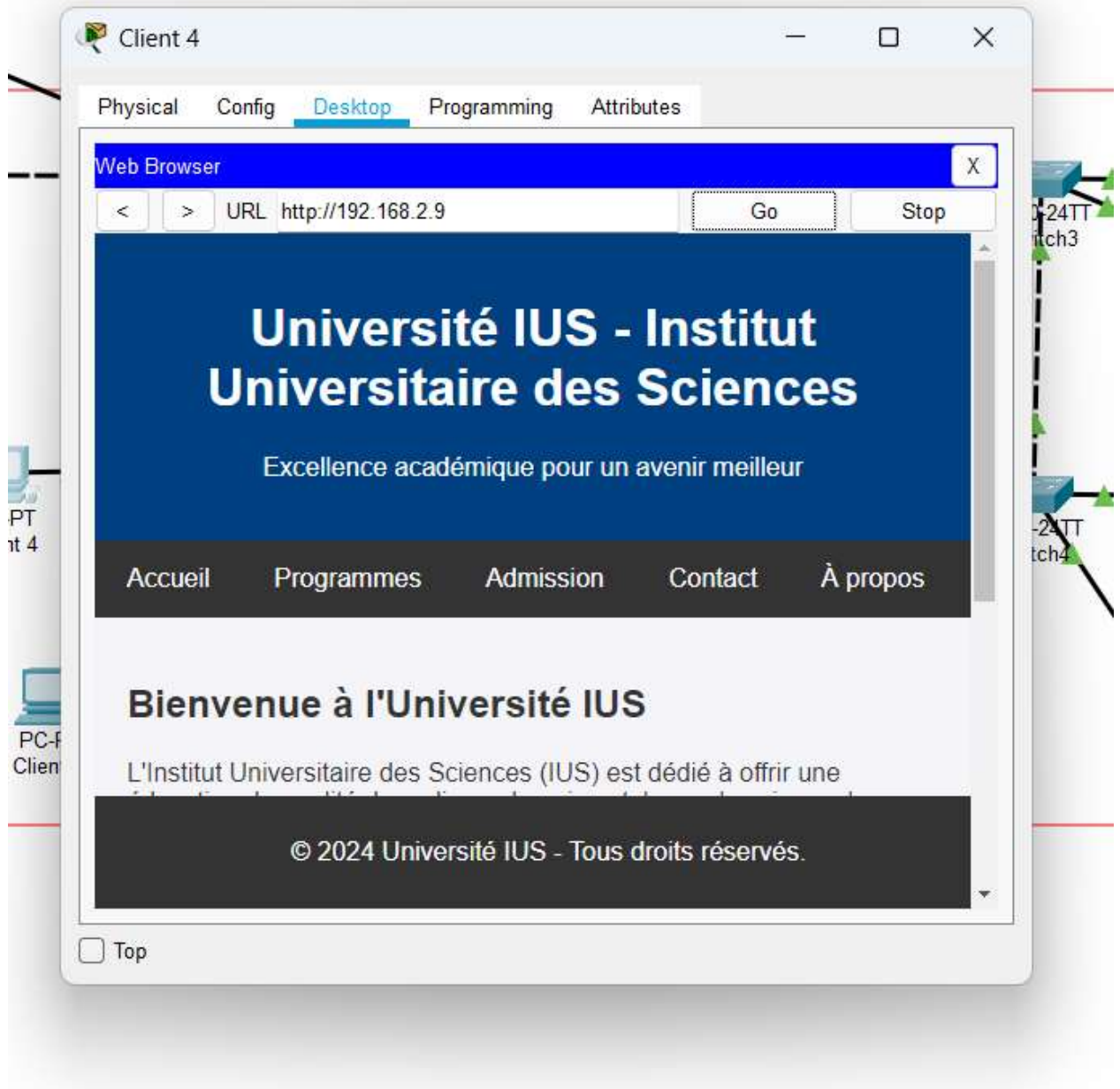
Radius EAP

File Name: copyrights.html

```
<html lang="fr">
<head>
<metacharset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Accueil - Université IUS</title>
<style>
/* Styles de base pour la page */
body {
font-family: Arial, sans-serif;

margin: 0;
padding: 0;
background-color: #f4f4f9;
color: #333;
}
header {
background-color: #004080;
color: white;
padding: 20px;
text-align: center;
}
nav {
background-color: #333;
color: white;
overflow: hidden;
}
nav a {
color: white;
padding: 15px 20px;
text-decoration: none;
float: left;
}
nav a:hover {
background-color: #555;
}
.container {
padding: 20px;
}
footer {
background-color: #333;
color: white;
text-align: center;
padding: 10px;
position: fixed;
bottom: 0;
width: 100%;
}
</style>
</head>
<body>
```

Depuis un Pc on ouvre le navigateur et tape l'address ip de notre server web.



## Conclusion

Ce travail m'a permis de savoir configurer un serveur DNS centralisé dans un réseau, associer plusieurs noms de domaine à différents serveurs (Web, Mail, FTP), configurer un serveur Web (HTTP et HTTPS) dans Cisco Packet Tracer, comprendre la différence entre HTTP et HTTPS. Donc la tâche a été réussie.